

Unicode, emoji, and the WinAnsi boundary

The 14 base PDF fonts use WinAnsi encoding, a single-byte mapping covering Latin-1 plus extra symbols (em-dash, curly quotes, ellipsis, currency, etc.). Text outside WinAnsi has no glyph in those fonts, so inkmd routes it three ways: **emoji render as color images** (from a bundled font); **Cyrillic, Greek, and Latin-Extended render through an embedded font**; and a codepoint no available font covers (e.g. CJK, which the bundled font lacks) shows a visible [U+XXXX] marker rather than a silent ?. This gallery shows what each path looks like.

Always available

ASCII printable:

!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[]^_`abcdefghijklmnopqrstuvwxyz{|}~

WinAnsi extras

Em dash: — (U+2014). En dash: – (U+2013). Curly quotes: "double" and 'single'. Ellipsis: ... (U+2026). Bullet: • (U+2022). Dagger: † (U+2020). Double dagger: ‡ (U+2021). Per mille: ‰ (U+2030). Trademark: ™. Registered: ®. Copyright: ©. Section: §. Paragraph: ¶.

Currency symbols in WinAnsi: \$ £ ¥ ¢ € €. Fractions: ½ ¼ ¾. Plus: ± × ÷.

Accented Latin: é è ê ë ñ ç ü ö ä Å Æ Ø ß ÿ.

Emoji (render as color images)

Single emoji: 🎉 🚀 ✅ ⚠️ 🔥 💡 📦. Skin tone: 👍. Flags: 🇯🇵 🇺🇸 🇬🇧. ZWJ sequences: 👨👩👦 (family) and 🏳️ (rainbow flag). Keycaps: 🔢 🔠 🔡 🔢.

Emoji are drawn from a bundled color font as small inline images scaled to the surrounding text size. They look crisp inline; at very large heading sizes the bitmaps soften slightly (a documented trade-off of the bitmap approach). The single-file zipapp build omits the font and falls back to a textual label such as [rocket].

Outside WinAnsi (embedded font or [U+XXXX] marker)

Cyrillic, Greek, and Latin-Extended render through the embedded font. A codepoint no

available font covers (CJK under the bundled font) shows a visible [U+XXXX] marker, and inkmd emits a MissingGlyphWarning. CJK full rendering is a planned later font pack.

Cyrillic: Привет, мир.

Greek: αβγδε ΑΒΓΔΕ, and a Δ (capital delta) by itself.

CJK (shows [U+XXXX] markers, the bundled font has no CJK):

[U+4F60][U+597D][U+4E16][U+754C] (Chinese),
[U+3053][U+3093][U+306B][U+3061][U+306F][U+4E16][U+754C] (Japanese hiragana),
[U+C548][U+B155][U+D558][U+C138][U+C694] (Korean).

Mathematical: $\sum \int \sqrt{\infty} \neq \leq \geq \in \forall \exists$.

Arrows: $\rightarrow \leftarrow \uparrow \downarrow \Rightarrow \Leftarrow \Leftrightarrow$ .

Mixed inline

A paragraph that mixes Latin, emoji, and other scripts: "Hello, bonjour, guten Tag, , Привет, [U+4F60][U+597D]". The Latin parts and the emoji render directly; the Cyrillic renders through the embedded font; and the CJK, which the bundled font does not cover, shows [U+XXXX] markers. inkmd splits the line per codepoint to route each one.

In code blocks

```
ASCII only: works.  
Em dash -: works (WinAnsi).  
Emoji   : renders as a color image.  
Delta Δ: renders via the embedded font.
```

```
A Python literal with Unicode:  
greeting = "Привет, мир"
```

In tables

Embedded-font routing does not yet reach table cells: the column widths are computed before the run split, so non-WinAnsi text inside a cell still renders ? (a known limitation, fix pending). The same text in prose renders through the embedded font, as the Cyrillic line above shows.

Region	Greeting	Status
English	Hello	OK
French	Bonjour	OK

Region	Greeting	Status
Emoji		renders
Russian	??????	?
Chinese	??	?